

Overview: Clues to Systemic Disease on Skin Examination

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- Overview: Clues to Systemic Disease on Skin Examination
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2025 Edition

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Chapter 1 Overview: Clues to Systemic Disease on Skin Examination

Overview: Clues to Systemic Disease on Skin Examination

The skin is a mirror of internal disease. Dermatological findings frequently provide the earliest and most visible clues to underlying systemic pathology. A thorough skin examination can reveal diagnoses ranging from autoimmune conditions and infections to malignancies and endocrine disorders. Recognizing these patterns is essential for early diagnosis and appropriate management.

Red Flag Clinical Features

Certain skin findings should immediately raise suspicion for systemic disease:

- Rash associated with joint pain, fever, weight loss, weakness, shortness of breath, or altered bowel function
- Rash not responding to topical treatment
- Erythema of the skin indicating inflammation around blood vessels
- Non-blanching palpable purplish fixed lesions that are painful and blistering — suggestive of vasculitis
- Unusual changes in pigmentation or texture of the skin
- Palpable dermal lesions secondary to granuloma, metastasis, or lymphoma

KEY TEACHING POINTS

The skin mirrors internal disease — always consider systemic causes when skin findings are atypical.

Red flag features include: rash with constitutional symptoms, non-blanching lesions, unusual pigmentation.

A rash that fails topical therapy is NOT treatment-resistant dermatitis — think systemic.

Non-blanching palpable purpura is a medical red flag — always investigate for vasculitis.

MCQ TRAPS & PITFALLS

Do NOT confuse blanching maculopapular rash (toxic erythema) with non-blanching palpable purpura (vasculitis).

A rash that fails topical therapy is NOT treatment-resistant dermatitis — consider systemic causes.

Palpable dermal nodules are NOT always benign cysts — consider granuloma, metastasis, and lymphoma.

Characteristic Rashes Indicating Underlying Systemic Disease

Several pathognomonic skin lesions are strongly associated with specific systemic conditions. Recognizing these patterns allows clinicians to initiate targeted investigations promptly.

Characteristic Rashes and Their Systemic Associations

Skin Lesion	Key Features	Systemic Associations
Erythema multiforme	Target lesions (3 concentric zones), acral distribution	HSV (most common trigger), Hepatitis B/C, Mycoplasma pneumoniae
Pyoderma gangrenosum	Rapidly progressive painful ulcer with violaceous undermined borders	IBD (UC > CD), Rheumatoid arthritis, Hematological malignancy

Skin Lesion	Key Features	Systemic Associations
Erythema nodosum	Tender, erythematous subcutaneous nodules on shins	Streptococcal infection, IBD, TB, Sarcoidosis, Behçet's disease
Vasculitis (cutaneous small-vessel)	Non-blanching palpable purpura, painful, may blister or ulcerate	Hepatitis B/C, SLE, Cryoglobulinemia, Lymphoma, Leukemia

1. Erythema Multiforme

Erythema multiforme (EM) is an acute, self-limited, and sometimes recurring skin condition considered to be a type IV hypersensitivity reaction. It is most commonly triggered by infections.

- **Most common trigger:** Herpes simplex virus (HSV)
- **Other associations:** Hepatitis B and C, Mycoplasma pneumoniae
- **Systemic symptoms** of the underlying infection typically precede the EM rash by 2–14 days
- **Epidemiology:** Can occur at any age; more common in young adults
- **Pathophysiology:** Type IV hypersensitivity reaction (cell-mediated immune response)

Presentation — Target Lesions ("Doll's Eye" Appearance):

- Classic "target" or "iris" lesions with three concentric zones of color change
- Central dusky red area (epidermal necrosis) → middle pale edematous ring → outer erythematous ring
- Acraly distributed (hands, feet, elbows, knees, extremities)
- Lesions evolve over 5–14 days
- May be pruritic or painful
- Mucosal involvement in approximately 70% of cases (oral most common)

Classification:

- **EM Minor:** Typical target lesions, acraly distributed, no mucosal involvement or with mild mucosal involvement only; generally well-appearing patient
- **EM Major:** Widespread lesions with significant mucosal involvement (oral, genital, conjunctival), associated systemic symptoms (fever, malaise)

Diagnosis: Clinical — based on characteristic target lesions and history; skin biopsy rarely needed but shows interface dermatitis with necrotic keratinocytes

Treatment: Treat underlying cause (e.g., acyclovir 400mg PO TID x 7–10 days for HSV); supportive care; topical corticosteroids for symptomatic relief; oral analgesics; mouthwash for oral lesions

Recurrence: Common, especially with HSV-associated EM (up to 30%); prophylactic acyclovir 400mg PO BID may be indicated for recurrent cases

Prognosis: Excellent — self-limited course over 2–4 weeks; EM major may take 4–6 weeks

OSCE PEARLS

When you see target lesions, immediately ask about recent HSV infection (history of cold sores).

EM minor vs major is distinguished by mucosal involvement and systemic symptom burden.

Prophylactic acyclovir is indicated for recurrent HSV-associated EM (≥6 episodes/year).

Always distinguish EM from Stevens-Johnson syndrome — SJS involves >10% body surface area with widespread epidermal sloughing.

2. Pyoderma Gangrenosum

Pyoderma gangrenosum (PG) is a rare, neutrophilic dermatosis characterized by rapidly progressive, painful ulcers. It is strongly associated with systemic disease and often overlooked in clinical settings.

Epidemiology: Female > Male; peak age 40–60 years; prevalence 3–10 per million

Associations:

Inflammatory bowel disease (ulcerative colitis > Crohn's disease)

Rheumatoid arthritis

Hematological malignancy (myelodysplastic syndrome, leukemia, monoclonal gammopathy)

Other: Hepatitis, Behçet's disease, ankylosing spondylitis, SLE

Presentation:

- Begins as a painful pustule or nodule that rapidly ulcerates
- Rapidly progressive painful ulcer with violaceous, undermined, ragged borders
- Base is necrotic with purulent discharge
- Surrounding skin is erythematous and indurated
- Lower extremities most common site (70% of cases)
- May also occur at surgical wound sites (pathergy)

Pathergy Phenomenon:

- New lesions may develop at sites of minor trauma including venipuncture sites, IV cannulation, and surgical incisions
- **Do NOT debride** — surgical intervention worsens the ulcer
- This is a critical clinical pearl — mismanagement is common

Diagnosis: Clinical; diagnosis of exclusion — must rule out infection (bacterial, fungal, mycobacterial), vasculitis, malignancy, and factitious ulceration

- Skin biopsy may show neutrophilic infiltrate but is non-specific and may be deferred if clinical picture is clear
- Inflammatory markers (ESR, CRP) may be elevated

Treatment:

- Systemic corticosteroids: prednisone 1 mg/kg/day (first-line)
- Cyclosporine 2–5 mg/kg/day (alternative or steroid-sparing)
- Other immunosuppressants: infliximab, adalimumab (anti-TNF), mycophenolate mofetil
- Topical: high-potency corticosteroids or tacrolimus for early/mild disease
- Treat underlying disease concurrently
- Wound care: gentle non-adherent dressings; compression if no arterial disease

IMPORTANT

Pyoderma gangrenosum is a diagnosis of exclusion — rule out infection, vasculitis, and malignancy first.

Always screen for IBD and hematological malignancy when pyoderma gangrenosum is diagnosed (FBC, blood film, anti-CCP, colonoscopy if indicated).

Pathergy means surgical debridement will worsen the ulcer — never debride these lesions without confirming the diagnosis.

New lesions appearing at IV cannula sites or biopsy sites should raise suspicion for PG in hospitalized patients.

3. Erythema Nodosum

Erythema nodosum (EN) is the most common form of panniculitis — inflammation of the subcutaneous fat tissue. It represents a delayed-type hypersensitivity reaction to a variety of antigenic stimuli.

- **Epidemiology:** Female > Male (3:1 to 6:1); peak age 20–30 years
- **Most common cause overall:** Idiopathic (up to 50% of cases)
- **Most common secondary cause:** Streptococcal pharyngitis
- **Seasonal:** More common in spring and autumn

Presentation:

- Acute onset of painful, tender subcutaneous nodules (1–5 cm, can reach up to 20 cm)
- Located predominantly on the bilateral pretibial regions (anterior shins)
- Not sharply ill-defined — often described as "poorly marginated"
- Very tender on palpation; exquisitely tender
- Overlying skin is erythematous, warm, and edematous
- Typically bilateral and symmetrical
- May be associated with fever, malaise, arthralgia, and ankle edema

Course:

- Spontaneous resolution occurs within 3–6 weeks without scarring
- Lesions Initially bright red → become purple/bruise-like (so-called "contusiform") → fade to yellow/brown before resolving entirely (color evolution mimics a bruise)
- Recurrence indicates underlying persistent cause — investigate thoroughly

Investigations:

- FBC (euphemia), CRP/ESR (elevated)
- Throat culture (Group A Streptococcus)
- ASO titers (anti-streptolysin O — elevated in recent streptococcal infection)
- CXR (hilar lymphadenopathy for sarcoidosis or TB)
- Stool cultures if GI symptoms (Yersinia)
- Tuberculin skin test or IGRA (QuantiFERON) if TB suspected
- Consider: serum ACE, fecal calprotectin, hepatitis serology

Treatment:

- Treat underlying cause
- Supportive: leg elevation, compression stockings, rest
- NSAIDs (ibuprofen 400–600mg PO TID or naproxen 250–500mg PO BID) for pain

- Potassium iodide 300mg PO TID (for persistent/recurrent cases)
- Systemic corticosteroids (prednisone 0.5–1 mg/kg/day tapering over 2–4 weeks) — only if severe or refractory; use cautiously as steroids may worsen underlying infection

KEY TEACHING POINTS

EN is the most common panniculitis — always think "What is driving the inflammation?"

Lofgren syndrome = EN + bilateral hilar lymphadenopathy + fever + arthralgia — a pathognomonic presentation of acute sarcoidosis.

The color evolution of EN mimics a bruise but is always exquisitely tender (a true bruise may not be).

Recurrent EN is a red flag for persistent underlying disease — do not stop investigating after one episode.

Major Causes of Erythema Nodosum

Cause Category	Specific Causes	Key Investigation
Infections	Streptococcal pharyngitis, TB, Mycoplasma, Yersinia, Mycobacterium leprae, Hepatitis B/C	Throat culture, CXR, Mantoux, Serology
Inflammatory bowel disease	Crohn's disease, Ulcerative colitis	Fecal calprotectin, Colonoscopy, CRP/ESR
Sarcoidosis	Lofgren syndrome (EN + bilateral hilar lymphadenopathy + fever + arthritis)	CXR, ACE levels, Serum calcium
Drugs	Sulfonamides, Oral contraceptive pills, Penicillins, NSAIDs	Drug history
Idiopathic	No cause identified (up to 50%)	Diagnosis of exclusion
Behçet's disease	Oral ulcers + genital ulcers + uveitis + EN	Pathergy test, Clinical criteria
Malignancy (rare)	Lymphoma (especially Hodgkin's), Leukemia	FBC, LDH, Imaging

4. Leukocytoclastic Vasculitis (Cutaneous Small-Vessel Vasculitis)

Cutaneous small-vessel vasculitis (CSVV) refers to inflammation and destruction of small blood vessels in the skin, predominantly postcapillary venules. The term "leukocytoclastic" refers to the nuclear debris (leukocytoclasia) seen from neutrophils.

- **Epidemiology:** Can occur at any age; more common in adults
- **Key feature:** Non-blanching palpable purpura — **always palpable** (distinguishes from simple petechiae)
- **Distribution:** Gravity-dependent areas — lower legs and ankles most commonly affected

Palpable Purpura — The Hallmark Sign:

- Non-blanching (does not fade with diascopy)
- Palpable (raised, can be felt on examination)

- Purplish-red, often with irregular borders
- May progress to hemorrhagic vesicles, bullae, or ulceration
- Painful (often described as burning or stinging)
- Most common on ankles and lower legs

Associations of CSVV:

- **Infections:** Hepatitis B (immune complex-mediated), Hepatitis C (cryoglobulinemia), Streptococcal infections, infective endocarditis, HIV
- **Autoimmune:** SLE, Rheumatoid arthritis, Sjögren syndrome
- **Malignancy:** Lymphoma, Leukemia (especially hairy cell leukemia)
- **Drugs:** Antibiotics (penicillins, sulfonamides), NSAIDs, ACE inhibitors, allopurinol
- **Cryoglobulinemia:** Type II (mixed monoclonal-polyclonal, often with Hepatitis C)
- **Henoch-Schönlein purpura (IgA vasculitis):** Involves IgA immune complex deposition; associated with joint pain, abdominal pain, and glomerulonephritis

IMPORTANT

Non-blanching palpable purpura on dependent areas is ALWAYS a vasculitis until proven otherwise — never dismiss this finding.

Always check urinalysis in any patient with palpable purpura — renal involvement changes management entirely.

Hepatitis B and C must be screened in EVERY new diagnosis of cutaneous vasculitis.

Skin biopsy should be taken from a lesion <48 hours old; older lesions become non-specific.

OSCE PEARLS

Perform diascopy (press a glass slide against the lesion) — vasculitic lesions do NOT blanch.

Palpable purpura + abdominal pain + arthritis + renal involvement = Henoch-Schönlein purpura (IgA vasculitis).

If a hospitalized patient develops new palpable purpura, immediately check: (1) urinalysis, (2) drug chart, (3) blood cultures.

HSP in children is often post-streptococcal and self-limited; in adults, screen aggressively for malignancy.

Steps in the Diagnosis of Vasculitis

Step	Investigation	Purpose
1	Skin biopsy (lesion <48 hours old)	Confirm leukocytoclastic vasculitis, assess vessel involvement
2	Urinalysis and renal function (U&Es, eGFR)	Screen for renal involvement (RPGN, nephritis)
3	FBC, CRP, ESR, LFTs	Assess systemic inflammation and organ function
4	Hepatitis B and C serology	Major secondary cause of vasculitis

Step	Investigation	Purpose
5	ANA, ANCA, Complement levels (C3, C4)	Screen for autoimmune/ANCA-associated disease
6	Serum protein electrophoresis, serum immunoglobulins	Screen for cryoglobulinemia and paraproteinemia
7	CXR	Screen for pulmonary involvement (sarcoidosis, infection)
8	Blood cultures and echo if indicated	Rule out infective endocarditis

Drug Eruptions and Their Systemic Implications

Drug-related skin eruptions are among the most common causes of rash in clinical practice. Recognizing the pattern of drug eruption is essential for timely drug withdrawal and appropriate management.

Common Drug Eruption Patterns

Rash Pattern	Typical Drugs	Description
Morbilliform (maculopapular)	Antibiotics (penicillins, sulfonamides), Allopurinol, Antiepileptics	Most common drug eruption; begins on trunk, spreads centrifugally; pruritic; blanching; typically appears 7–10 days after drug initiation
Urticaria	NSAIDs, Penicillins, Opiates, ACE inhibitors	Wheals with central pallor; intensely pruritic; angioedema if deeper involvement
Fixed drug eruption	Trimethoprim, NSAIDs, Tetracyclines	Recurring round erythematous plaque at same site with each drug exposure; may become hyperpigmented
Erythema multiforme-like	NSAIDs, Sulfonamides, Allopurinol	Target-like lesions; must distinguish from infectious EM
SJS/TEN	Allopurinol, Antiepileptics, Sulfonamides	Life-threatening; widespread epidermal detachment; mucosal involvement; high mortality (TEN: 25–35%)

Warning Signs of Severe Cutaneous Adverse Reactions (SCARs)

Severe cutaneous adverse reactions represent a spectrum of potentially life-threatening drug reactions that require immediate recognition and management.

Stevens-Johnson Syndrome (SJS): Epidermal detachment <10% body surface area; severe mucosal involvement (oral, ocular, genital); prodrome of fever and malaise; Nikolsky sign positive (lateral pressure causes epidermal detachment)

Toxic Epidermal Necrolysis (TEN): Epidermal detachment >30% BSA; same as SJS but much more extensive; mortality up to 25–35%; often drug-induced

overlap SJS-TEN: Detachment 10–30% BSA

DRESS (Drug Reaction with Eosinophilia and Systemic Symptoms): maculopapular rash + eosinophilia + facial edema + hepatitis (most common organ involvement) + lymphadenopathy; onset typically 2–8 weeks after drug initiation; associated with allopurinol, antiepileptics, sulfonamides

KEY TEACHING POINTS

DRESS syndrome has a late onset (2–8 weeks) — do NOT dismiss a new rash as unrelated to a drug started 1 month ago.

Always calculate the body surface area of epidermal detachment in SJS/TEN — this determines management pathway and ICU referral.

Allopurinol is the most common cause of both DRESS and TEN — review allopurinol use whenever these conditions are suspected.

Eosinophilia on FBC in a patient with rash and recent drug exposure = think DRESS.

Skin Manifestations of Endocrine Disease

The skin is frequently affected in endocrine disorders, and cutaneous manifestations may precede the diagnosis of systemic disease.

Acanthosis Nigricans and Insulin Resistance

Acanthosis nigricans (AN) is characterized by hyperpigmented, velvety thickening of the skin, most commonly affecting body flexures.

- **Presentation:** Symmetrical, velvety, brown-black hyperpigmentation; most common sites are axillae, groin, neck, and antecubital fossae
- **Benign (metabolic) AN:** Strongly associated with insulin resistance, obesity, type 2 diabetes mellitus, and polycystic ovarian syndrome (PCOS); the most common type
- **Malignant AN:** Associated with internal malignancy, most commonly gastric adenocarcinoma; onset is rapid, extensive, and may involve mucosal surfaces (oral mucosa, lips)
- **Rule of thumb:** New-onset extensive AN in a thin, non-diabetic adult — urgently investigate for malignancy

Clinical Significance:

- In children and adolescents, AN is a clinical marker of insulin resistance and increased risk of type 2 diabetes — screen with fasting glucose, HbA1c, and lipid panel
- The severity of AN correlates with the degree of insulin resistance
- In malignant AN, the triad includes: tripe palms (velvety palms), acanthosis nigricans, and multiple skin tags

Pretibial Myxedema (Thyroid Dermopathy)

Pretibial myxedema is a skin manifestation of Graves' disease, resulting from deposition of glycosaminoglycans in the dermis.

Presentation: Non-pitting edema of the bilateral pretibial regions; skin is waxy, firm, indurated with a "peau d'orange" appearance; may extend to dorsum of feet

Associations: Graves' disease (most common cause of hyperthyroidism); typically occurs in patients with Graves' ophthalmopathy (orbitopathy); may also occur in hypothyroidism

Histopathology: Mucin (hyaluronic acid) deposition in the dermis

Treatment: Treat the underlying thyroid disorder; topical high-potency corticosteroids (e.g., clobetasol propionate 0.05% ointment) under occlusion for mild cases; severe cases may require IVIG or rituximab

Skin Signs of Common Endocrine Disorders

Skin Finding	Endocrine Disorder	Key Associations
Acanthosis nigricans (benign)	Type 2 diabetes, PCOS, Insulin resistance, Metabolic syndrome	Obesity; screen fasting glucose, HbA1c, lipids
Acanthosis nigricans (malignant)	Gastric adenocarcinoma most common	Rapid onset, extensive, mucosal involvement, weight loss, tripe palms
Pretibial myxedema	Graves' disease (hyperthyroidism)	Non-pitting pretibial edema, peau d'orange, Graves' ophthalmopathy
Xanthomas	Hyperlipidemia, Familial hypercholesterolemia	Tendon xanthomata (Achilles, extensor tendons), palmar xanthomata, eruptive xanthomata
Skin tags	Insulin resistance, Acromegaly, Bartsocas-Papadopoulos syndrome	Benign fibromas; may indicate metabolic risk

Dermatological Clues to Internal Malignancy

Several skin lesions serve as paraneoplastic markers of internal malignancy. Recognition of these signs can lead to earlier cancer diagnosis and improved outcomes.

Acanthosis nigricans (malignant type): Gastric adenocarcinoma most common; also breast, lung, and other GI malignancies; look for tripe palms, sudden onset in non-obese patient

Dermatomyositis: Heliotrope rash (purple discoloration of eyelids) + Gottron's papules (violaceous papules over knuckles) + proximal muscle weakness; in adults >40 years, ~20–30% have an associated malignancy (most commonly ovarian, lung, GI, lymphoma)

Acanthosis palmaris (tripe palms): Velvety thickening of palmar surfaces; associated with lung cancer (most common) and gastric cancer; often coexists with malignant AN

Erythema gyratum repens: Concentric, wood-grain-like erythematous bands that migrate across the skin at >1 cm/day; strongly associated with underlying malignancy (lung most common); present before cancer diagnosis in ~80% of cases

Leser-Trélat sign: Sudden eruption of multiple seborrheic keratoses; associated with adenocarcinoma (stomach, colon, breast) and lymphoma (especially mycosis fungoides)

Necrolytic migratory erythema: Serpiginous, erythematous, erosive rash with crusted borders; classically associated with glucagonoma (pancreatic alpha-cell tumor); part of the "4D syndrome" (dermatitis, depression, DVT, diabetes)

Sweet syndrome (acute febrile neutrophilic dermatosis): Tender, erythematous plaques with pseudovesiculation; associated with acute myeloid leukemia (AML), myelodysplastic syndrome, and solid tumors; note the neutrophilic dermal infiltrate on biopsy

IMPORTANT

In a patient over 40 years presenting with new dermatomyositis, ALWAYS screen for occult malignancy: CT chest/abdomen/pelvis, tumor markers (CA-125, CEA), age-appropriate cancer screening, and colonoscopy if not recently performed.

Erythema gyratum repens is nearly pathognomonic for malignancy — investigate aggressively even if the patient is asymptomatic.

Sweet syndrome can be the first presentation of AML — a new unexplained neutrophilic dermatosis in an adult warrants FBC with differential and peripheral blood film.

Skin Manifestations of Hepatobiliary Disease

The skin provides important diagnostic clues to liver and biliary disease. Many of these findings reflect impaired hepatic synthesis and metabolism.

Chronic Liver Disease

Jaundice: Yellow discoloration of skin and sclerae due to elevated bilirubin (>2.5 – 3.0 mg/dL or ~ 40 – 50 $\mu\text{mol/L}$); first visible in sclerae because of high elastin content binding bilirubin

Spider naevi (spider angiomas): Central arteriole with radiating capillaries; blanch from center outward; most common on upper trunk, face, and arms; associated with elevated estrogen levels due to impaired hepatic estrogen metabolism; >3 spider naevi is abnormal

Palmer erythema: Diffuse erythema of the thenar and hypothenar eminences; due to local vasodilation and elevated estrogen; present in approximately 23% of cirrhotic patients

Caput medusae: Dilated periumbilical veins indicating portal hypertension with portosystemic shunting; blood flows away from the umbilicus (distinguishing from IVC obstruction where flow is upward)

Pruritus: Common in cholestatic liver disease (PBC, PSC); due to bile salt deposition in skin; generalized, worse at night

KEY TEACHING POINTS

Spider naevi >3 are pathognomonic for liver disease — count them at every abdominal examination.

Palmar erythema in a patient with jaundice = think chronic liver disease, but also consider physiological (pregnancy, up to 30% of normal population), rheumatoid arthritis, and thyrotoxicosis.

Caput medusae = portal hypertension; recanalized umbilical vein; direction of blood flow distinguishes from SVC/IVC obstruction.

Generalized pruritus without primary skin lesions in an elderly woman = think Primary Biliary Cholangitis (PBC).

Practice Questions

Q: A 28-year-old woman presents with a 3-day history of painful, raised, purplish lesions on her lower legs that do not blanch on pressure. She also reports joint pain, fever, weight loss, and malaise. Urinalysis shows hematuria and proteinuria. Blood tests reveal elevated CRP and ESR. What is the most likely diagnosis?

- A. Allergic contact dermatitis
- B. Erythema nodosum
- C. Leukocytoclastic vasculitis
- D. Cellulitis
- E. Psoriasis

Answer: C

Explanation: The key features pointing to leukocytoclastic vasculitis are: (1) non-blanching palpable purpura (the hallmark); (2) systemic symptoms (fever, malaise, joint pain); (3) renal involvement (hematuria and proteinuria — systemic vasculitis not limited to skin). Erythema nodosum presents with tender subcutaneous nodules on the shins, not purpura. Cellulitis would be erythematous and warm but typically unilateral and without renal involvement.

Q: A 19-year-old college student develops target-like lesions on the palms and soles after recovering from a cold sore 10 days ago. The lesions have three concentric zones of color change. She has mild oral mucosal erosions but no systemic symptoms. Which is the most likely trigger for her presentation?

- A. Mycoplasma pneumoniae
- B. Hepatitis B
- C. Herpes simplex virus
- D. Drug reaction
- E. Streptococcal infection

Answer: C

Explanation: Herpes simplex virus (HSV) is the most common trigger for erythema multiforme, accounting for >50% of cases. The scenario describes EM with typical target lesions and mild mucosal involvement (EM minor). The preceding history of a cold sore (HSV reactivation) 2–14 days before the rash is classic. While Mycoplasma pneumoniae and Hepatitis B are also associated with EM, HSV is the most common cause and the temporal relationship with the cold sore makes it the clear answer.

Q: A 55-year-old woman with a history of ulcerative colitis presents with a rapidly enlarging, exquisitely painful ulcer on her left lower leg. The ulcer has violaceous, undermined borders and a necrotic base. The patient notes that a similar lesion appeared at an IV cannula site during a recent hospital admission. Platelet count and inflammatory markers are elevated. Which of the following is the most appropriate next step in management?

- A. Surgical debridement of the ulcer
- B. Broad-spectrum IV antibiotics
- C. Skin biopsy followed by initiation of systemic corticosteroids
- D. Compression therapy alone
- E. Topical antibiotic ointment

Answer: C

Explanation: This presentation is classic for pyoderma gangrenosum: rapidly progressive painful ulcer with violaceous undermined borders in a patient with IBD, with pathergy (lesion developing at IV cannula site). The most appropriate next step is to confirm the diagnosis (skin biopsy to exclude mimics) and initiate systemic immunosuppression (corticosteroids). Surgical debridement (Option A) is contraindicated due to pathergy — it would worsen the ulcer. Antibiotics alone are insufficient as PG is an immune-mediated condition.

Q: A 25-year-old woman presents with bilateral, exquisitely tender, erythematous nodules on her anterior shins. She reports a sore throat 2 weeks ago. CXR is normal. FBC shows mild leukocytosis, CRP is elevated, and ASO titers are elevated. What is the most likely underlying cause?

- A. Sarcoidosis
- B. Streptococcal infection
- C. Tuberculosis
- D. Inflammatory bowel disease
- E. Behçet's disease

Answer: B

Explanation: This presentation is classic erythema nodosum. The preceding sore throat with elevated ASO titers strongly suggests streptococcal pharyngitis as the underlying cause — the most common identifiable trigger of EN. Normal CXR argues against sarcoidosis and TB. While IBD and Behçet's disease can cause EN, there is no GI or mucosal ulcer history to suggest these. Treatment is supportive: NSAIDs for pain, leg elevation, and treating the underlying infection.

Q: A 62-year-old non-obese, non-diabetic man presents with a 2-month history of rapidly progressive, dark, velvety thickening of the skin in his axillae, groin, and around his neck. He reports a 7-kg unintentional weight loss over 3 months. Physical examination also reveals multiple skin tags and velvety plaques on his palms. Which malignancy should be urgently investigated?

- A. Non-small cell lung cancer
- B. Gastric adenocarcinoma
- C. Colon adenocarcinoma
- D. Prostate cancer
- E. Hepatocellular carcinoma

Answer: B

Explanation: Malignant acanthosis nigricans is most commonly associated with gastric adenocarcinoma. The key clues here are: (1) rapid-onset, extensive AN in a non-obese, non-diabetic adult (rules out benign metabolic AN); (2) tripe palms (velvety palmar thickening); (3) multiple skin tags; (4) unintentional weight loss. This triad (AN + tripe palms + weight loss) demands urgent upper GI investigation — esophagogastroduodenoscopy (OGD) and CT abdomen.

Q: A 45-year-old woman is admitted with a widespread skin rash, fever, facial swelling, and jaundice. She was started on allopurinol for gout 6 weeks ago. FBC shows marked eosinophilia and atypical lymphocytes. LFTs are deranged (ALT 450 U/L). What is the most likely diagnosis?

- A. Erythema multiforme

- B. Stevens-Johnson syndrome
- C. Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS)
- D. Acute viral hepatitis
- E. Toxic epidermal necrolysis

Answer: C

Explanation: This is a classic presentation of DRESS syndrome: (1) delayed-onset rash 2–8 weeks after drug initiation (allopurinol started 6 weeks ago); (2) facial edema; (3) eosinophilia on FBC; (4) hepatic involvement (elevated ALT, jaundice); (5) systemic symptoms (fever). Allopurinol is the most common cause of DRESS. The hallmark combination of rash + eosinophilia + organ involvement (hepatic most common) + delayed onset is characteristic. SJS/TEN presents much earlier (within days) and with epidermal detachment, not eosinophilia.

Q: A 38-year-old woman presents with a widespread maculopapular rash that started on her trunk and has spread to her limbs. She was started on amoxicillin for a dental infection 10 days ago. The rash is pruritic and blanches completely on diascopy. She is afebrile with no mucosal involvement. What is the most appropriate management?

- A. Continue the antibiotic and add oral antihistamines
- B. Stop the antibiotic and prescribe oral antihistamines and topical emollients
- C. Admit for IV corticosteroids
- D. Perform a skin biopsy
- E. Prescribe oral acyclovir

Answer: B

Explanation: This is a classic morbilliform (maculopapular) drug eruption — the most common drug rash, caused by amoxicillin, appearing 7–10 days after drug initiation. Key features: blanching maculopapular rash, centrifugal spread (trunk to limbs), no mucosal involvement, no systemic symptoms. Management is straightforward: stop the offending drug, provide symptomatic relief with oral antihistamines and topical emollients. The rash will self-resolve within 1–2 weeks of drug withdrawal. No biopsy or steroids are needed in an uncomplicated case.